5th Year Critical Angle Questions: Worked Solutions

The 5-step method

Every question below is solved in the same five steps, and critical angles are rounded only at the final step.

1. List all known quantities with values and SI units.
2. Write down what is being asked for.
3. Select the equation that links the knowns and the unknown, and state why it was chosen.
4. Substitute values with units and solve.
5. Check: does the magnitude make physical sense, and are the units correct?

Question 1

The critical angle of a type of plastic is 40°. Find the refractive index.

Step 1: Knowns

- Critical angle, C = 40°
- Refractive index of air, n(air) = 1 (the light leaves the plastic into air)

Step 2: Unknown

- The refractive index of the plastic, n

Step 3: Equation

$\sin C = 1/n$

This equation links the critical angle directly to the refractive index of the denser medium, and these are the only two quantities involved.

Step 4: Substitute and solve

$n = 1 / \sin 40° = 1 / 0.6428 = 1.56$

Step 5: Check

This is greater than 1 and typical of plastics (about 1.4 to 1.6). It has no units, as refractive index is a ratio.

Question 2

Sapphire (Al_2O_3) has a high refractive index of 1.77. What is the critical angle for sapphire?

Step 1: Knowns

- n(sapphire) = 1.77
- Surrounding medium: air, n = 1

Step 2: Unknown

- The critical angle, C

Step 3: Equation

sin C = 1/n

We know n and want C, so this is the direct link.

Step 4: Substitute and solve

sin C = 1 / 1.77 = 0.5650 ⇒ C = $\sin^{-1}(0.5650)$ = 34.4°

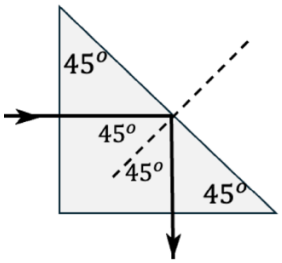
Step 5: Check

The critical angle is smaller than that of the plastic in Question 1, which is correct because sapphire has a higher refractive index. The answer is in degrees, as expected for an angle.

Question 3

What is the smallest value of the refractive index that allows total internal reflection in the 45° prism?

Step 1: Knowns



- The ray enters the vertical face along the normal, so it is not deviated.
- It hits the sloping face (hypotenuse) at an angle of incidence of 45°.
- For total internal reflection to just occur, C = 45°.
- Outside the prism is air, n = 1.

Step 2: Unknown

- The smallest refractive index n that allows total internal reflection

Step 3: Equation

sin C = 1/n

Total internal reflection requires the angle of incidence to be at least the critical angle. The smallest refractive index corresponds to the largest critical angle that still works, which is 45°.

Step 4: Substitute and solve

n = 1 / sin 45° = 1 / 0.7071 = 1.41

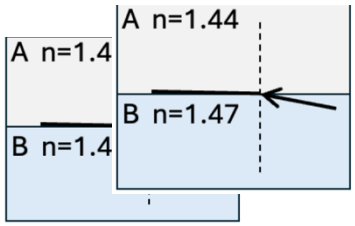
Step 5: Check

Any material with n ≥ 1.41 has C ≤ 45°, so the 45° ray is totally internally reflected. Glass (n ≈ 1.5) works, which is why glass prisms are used this way in binoculars and periscopes.

Question 5

Plastic A and Plastic B are in contact. Their refractive indices are 1.44 and 1.47 respectively. What is the critical angle going from B to A?

Step 1: Knowns



- n(A) = 1.44
- n(B) = 1.47
- The light travels from B to A.

Step 2: Unknown

- The critical angle for light going from B into A

Step 3: Equation

$\sin C = n_2/n_1$

Here n1 is the refractive index of the medium the light starts in (B) and n2 is that of the medium it enters (A). This is the general form of sin C = 1/n, where n2 = 1 for air. Light is going from the higher to the lower index, so a critical angle exists.

Step 4: Substitute and solve

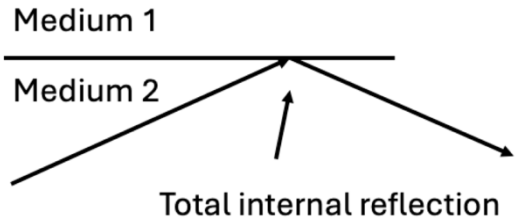
$\sin C = 1.44 / 1.47 = 0.9796 \Rightarrow C = \sin^{-1}(0.9796) = 78.4°$

Step 5: Check

The two indices are very close, so the critical angle should be close to 90°, and 78.4° fits. The ratio is less than 1, so the inverse sine is valid.

Question 6

Which medium has the highest refractive index? (The diagram shows total internal reflection at the boundary between Medium 1 and Medium 2.)



Step 1: Knowns

- The diagram shows total internal reflection at the boundary, with the ray travelling in Medium 2 and reflecting back into Medium 2.

Step 2: Unknown

- Which medium has the highest refractive index

Step 3: Principle

Total internal reflection can only happen when light travels from a medium of higher refractive index towards one of lower refractive index. This is the condition for a critical angle to exist, since sin C = n2/n1 requires n2 < n1.

Step 4: Reasoning

The ray is incident from Medium 2 onto the boundary with Medium 1 and is totally internally reflected. Therefore light was going from Medium 2 towards Medium 1, so n2 > n1.

Step 5: Check

If Medium 1 were denser, light entering it would bend towards the normal and could not be totally internally reflected. The diagram is consistent with **Medium 2 having the highest refractive index**.